

A Transparency Checklist for Theory-Driven Work in Psychology

Version: 5.0 — *this is a living document and subject to updates*

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In light of the Replication Crisis and Open Science Movement (Open Science Collaboration, 2015; Tyner et al., 2026), transparency in research has been increasingly recognized as being of key importance for rigorous scientific practice. Numerous standards and checklists have been developed to improve transparency in reporting (e.g., Aczel et al., 2019; Schubert et al., 2025). Yet, most of the available resources focus on methodological and statistical aspects of reporting, leaving the quality of theory development and testing largely unaddressed. Theory development and testing, however, are essential parts of cumulative science, and a lack of proper theorizing in psychological research has been identified as a main driver of its low replicability (e.g., Eronen & Bringmann, 2021; Muthukrishna & Henrich, 2019; Oberauer & Lewandowsky, 2019; Van Dongen et al., 2026). Critics specifically highlight the ambiguity and low precision in psychological theories, resulting in a weak link between a theory's predictions and certain empirical observations (Frankenhuis et al., 2023; Glöckner et al., 2026; Oberauer & Lewandowsky, 2019; Van Dongen et al., 2026). To help improve identified weaknesses, we introduce a transparency checklist for theory-driven work in psychology and adjacent behavioral sciences.

Aims of the Checklist

The checklist serves two complementary aims. First, it can be used as a *reporting guideline* for researchers who engage in theory-driven work. By specifying which aspects of a theory, its formalization, and its evaluation should be documented, the checklist helps authors produce more transparent, complete, and reproducible accounts of their theoretical contributions (Glöckner et al., 2026). Second, the checklist can be used as an *assessment tool* for evaluating the theoretical rigor of published or submitted work – for instance, in the context of peer review or research assessments for hiring and promotion (Gärtner et al., 2025; Schönbrodt et al., 2025). In this way, the checklist can both facilitate and incentivize theory-driven work in psychology (Lange et al., 2025).

Structure of the Checklist

The checklist is organized around five key areas of theory-driven work: 1) articulating a theory and its core elements verbally, 2) formalizing a theory in a formal model, and evaluating a theory/model through 3) simulations, 4) empirical data, and/or 5) comparison against other theories/models. For each area, the checklist specifies concrete indicators that a reader (or author) can check to assess whether the relevant aspects of the theoretical work have been reported transparently. Importantly, the checklist is modular and does not require every work to cover all five areas. Filter items allow researchers to indicate which areas are relevant to their particular contribution, so that only the applicable indicators need to be addressed.

Scope of the Checklist

While the checklist is aimed to be applicable to a broad range of theoretical and theory-driven work in psychology and adjacent behavioral sciences, it is characterized by certain philosophical and epistemological principles. The checklist incentivizes explanation-oriented theoretical work that aims to explain phenomena by specifying the mechanisms that give rise to them (Glennan, 2017; Machamer et al., 2000; Van Dongen et al., 2025). Furthermore, it values formalization as a way to increase the precision of theoretical claims (Guest & Martin, 2021; Robinaugh et al., 2021; Van Dongen et al., 2025). As such, the checklist is not applicable to purely descriptive or exploratory

research and may be less applicable to work grounded in radically different epistemological traditions, or to certain forms of psychometric or measurement-focused work that do not involve substantive explanatory theory. That said, many of the reporting standards articulated here reflect general principles of good scientific practice and may be useful beyond the specific philosophical framework that informed their development.

In the following, we first introduce the terminology that defines the key terms used throughout the checklist and then present the checklist itself.

Terminology

Table 1 summarizes the core terms used in the checklist. We define a *theory* as a set of interrelated propositions about aspects of the world that aim to explain and predict certain phenomena (Guest & Martin, 2021; Van Dongen et al., 2025). Theories typically do so by specifying *mechanisms*—the causal or functional pathway through which entities, processes, and their organization give rise to relevant phenomena. We define *phenomena* as stable features of the world that may constitute the targets of scientific explanation. In psychology, such features are typically inferred from robust patterns of *observations*, that is, observations that are made repeatedly and through the use of different procedures (Bogen & Woodward, 1988). Notably, the assumption of the existence of the phenomenon may be false (e.g., due to errors in measurement).

Theories typically consist of *concepts* (i.e., abstract ideas used to describe, differentiate, or reason about certain aspects of the world), relations among these concepts, and further assumptions. Together, we call these the *elements of a theory*.

Our terminology further adopts a semantic view of scientific theories. We assume that a theory is always expressed in some verbal form (i.e., in natural language) to communicate it to others, but it may also be expressed in, for example, a formal model. A *formal model* represents a specific version of (parts of) a theory expressed through formal means such as mathematical equations, algorithms, or logical propositions (Glöckner & Fiedler, 2025; Guest & Martin, 2021; Robinaugh et al., 2021; Smaldino, 2020; Van Rooij & Blokpoel, 2020).

If a theory is expressed in a formal model, then, the elements of the theory are typically represented in the formal model by variables, parameters, constants, and the functions and operators that connect them. Together with any auxiliary assumptions added during formalization, these form the *elements of a formal model*. The same verbal theory may be expressed in different formal models.

Theories may be *evaluated* via simulations, empirical data, and/or comparisons to other theories (Guest, 2024; Meehl, 1990). Running *simulations* always requires that the theory is expressed in a formal model and enables the generation of concrete predictions under an assumed set of conditions. This can be used to evaluate if a theory/model is able to reproduce data patterns that are indicative of relevant phenomena that the theory aims to explain (Van Dongen et al., 2025). Evaluations by empirical data do not necessarily require a theory to be expressed in a formal model but may be carried out on purely verbal theories. Testing a theory empirically, however, typically requires specifying how each element of the theory that is relevant to the empirical test has been *operationalized* (i.e., how it has been measured, manipulated, or estimated). Evaluating only *parts* of a theory is possible and legitimate.

Expressing a theory in a formal model, simulating from it, and testing it empirically often requires making *auxiliary assumptions* (e.g., about distributions of variables, parameter ranges, or measurement procedures), which are not directly implied by the theory itself. Such auxiliary assumptions need to be made explicit, to distinguish them from the core of the theory.

Table 1

Definitions of core terms used in the checklist.

Term	Definition
Theory	A set of interrelated propositions about aspects of the world that aim to explain and predict certain phenomena.
Phenomena	Stable features of the world that may constitute the targets of scientific explanation. In psychology, such features are typically inferred from robust patterns of observations (i.e., observations that are made repeatedly and through the use of different procedures).
Concept	An abstract idea used to describe, differentiate, or reason about certain aspects of the world. The same concept may be denoted using different words and the same word may be used to denote different concepts.
Elements of a Theory	The constituent building blocks of a theory. That is, the concepts, their relations, and any other core assumptions that together form those interrelated propositions about aspects of the world that constitute the theory.
Mechanism	The causal or functional pathway through which entities, processes, and their organization give rise to a phenomenon.
Formal Model	A specific version of (a part of) a theory expressed through formal means, such as mathematical equations, algorithms, or logical propositions.
Elements of a Formal Model	All the variables, parameters, constants, functions, operators, and auxiliary assumptions that were used to arrive at a specific formalization of a theory.
Auxiliary Assumption	Any assumption or decision not directly implied by the theory but introduced for purposes such as formalization, simulation, prediction, or empirical testing.

Observation	A record of a state, event, or behavior. Observations can be empirical (i.e., obtained through sensory or instrumental means) or simulated (i.e., the outcome of a simulation from a formal model). In scientific practice, observations are typically recorded and structured as data .
Prediction	An expected pattern of observations derived from a theory or formal model under a set of assumed conditions (including auxiliary assumptions).
Simulation	The process of running a formal model forward to observe its behavior under an assumed set of conditions.
Operationalization	The process of specifying how (some of) the elements of a theory/formal model are measured, manipulated, or estimated to make the theory testable.

The Checklist

No.	Question / Item	Explanation
Filter (T)	☐ This research was mainly descriptive / exploratory, without a clear theoretical rationale > out ☐ This research was theoretically motivated and (further) developed AND / OR evaluated at least one (part of) a theory > T-Items ☐ Cannot answer the question > out	These criteria apply not only to purely theoretical work but also to original empirical work that incorporates theoretical reasoning in generating its hypotheses. Thus, if the paper states any theoretically derived hypotheses or the like, you should choose the second option here. Orienting Question: <i>Was there a theoretical motivation for conducting this research?</i>
T...	<i>The paper...</i>	The following criteria also apply when reusing or testing a previously developed theory / formal model. In this case (some of) the points can be earned if the required information is provided in a clear reference to that previous work. However, any deviations, additions, and adaptations should be reported in the current work.
T1	☐ ...contains an explicit verbal description of at least one phenomenon to be explained by the developed / tested theory.	<i>Phenomena</i> are stable features of the world that may constitute the targets of scientific explanation. In psychology, such features are typically inferred from robust patterns of observations (i.e., observations that are made repeatedly and through the use of different procedures). Orienting Question: <i>Does it become clear what the theory aims to explain?</i>
T2	☐ ...provides a working definition for all central concepts used in the theory or provides a clear reference to a previous definition of the concept.	A <i>concept</i> is an abstract idea used to describe, differentiate, or reason about certain aspects of the world. Concepts might not always have an accepted definition. Yet, it is important to provide a working definition and to clearly communicate how the concept is used / understood in the context of the theory. Orienting Question: <i>Is it clear what the concepts mentioned in the theory are about?</i>

T3	☐ ...defines (in verbal form) the central relationships among the concepts used in the theory or provides a clear reference to a previous definition of this relationship.	Verbal definitions are still required even if concepts and their relations are expressed in a formalized way (e.g., in a formula). While formalizations are inherently more precise, verbal expressions are still necessary for others to properly understand the content of the formalization. Orienting Question: <i>Is it clear how the concepts mentioned in the theory are related to each other?</i>
T4	☐ ... describes a mechanism explaining how the relevant phenomena arise.	A <i>mechanism</i> is the causal or functional pathway through which entities, processes, and their organization give rise to a phenomenon. Orienting Question: <i>Is it clear how the theory explains the phenomenon?</i>
T5	☐ ...discusses the intended scope of the theory and, where possible, specifies known boundary conditions or limitations of its applicability.	This concerns assumptions about generality (e.g., subject and stimulus population characteristics) as well as simplifying assumptions. When building on an existing theory, a clear reference to a prior scope statement can be sufficient if the scope is unchanged. Orienting Question: <i>Is it clear under which conditions / in which situations the theory is applicable?</i>
Filter (F)	☐ ... <i>formalizes</i> (parts of) the theory in a formal model. That is, the properties of and the connections among the elements of the theory are expressed in a formalized way > F-Items ☐ ... <i>does not formalize</i> (parts of) the theory in a formal model > out	A <i>formal model</i> expresses (a part of) a theory through formal means, such as mathematical equations, algorithms, or logical propositions. It should be noted that a formal model specifies a data generating process that enables the running of simulations. This is different from a statistical model, which mainly aims to describe data instead of explaining its generating process.

F...	The paper...	
F1	☐ ...specifies any auxiliary assumption that had to be added to enable the translation of the verbal theory into a formal model.	An <i>auxiliary assumption</i> is any assumption or decision not directly implied by the theory but introduced for purposes such as formalization, simulation, prediction, or empirical testing. Orienting Question: <i>Are modelling decisions that are not directly implied by the theory made explicit?</i>
F2	☐ ...lists all elements of the formal model and defines them by connecting them to their corresponding elements of the verbal theory (if available). If an element of the model represents an auxiliary assumption, this is made explicit.	<i>Elements of the formal model</i> subsume all the variables, parameters, constants, functions, operators, and auxiliary assumptions that were used to arrive at a specific formalization of a theory. Specifying all the elements of a formal model can also be achieved using a Table or an Appendix. Common mathematical operators can be exempt from this requirement. Orienting Question: <i>Is it clear what each element in the formal model is supposed to represent?</i>
F3	☐ ...specifies the basic mathematical properties of all elements of the formal model (e.g., scalar vs. matrix, dimensionality, etc.).	Even if all elements of the formal model have a narrative definition, their mathematical properties might be unclear (e.g., is parameter <i>B</i> a constant, a vector, or a two-dimensional matrix?). This can also be ensured by referring to an established standard notation. Orienting Question: <i>Are the (mathematical) properties for each element of the model clear?</i>
F4	☐ ...specifies constraints on the possible values for all elements of the formal model (e.g., upper and lower bounds, scale level, prior distributions). If a parameter has no assumed constraints/scale, this is made explicit.	These constraints may be directly implied by the theory, introduced through auxiliary assumptions, or based on empirical observations. Orienting Question: <i>Is it clear which values each element in the model can possibly take?</i>

F5	☐ ...states whether the model is structurally identifiable — that is, whether distinct parameter values produce distinct patterns of predictions — or defines constraints under which identifiability can be achieved. If identifiability does not apply to the model or was not assessed, this is stated explicitly.	<i>Structural identifiability</i> is a property of the model, not of any single element. It concerns whether the model's parameters can, in principle, be distinguished from the full pattern of predictions the model generates. A model is structurally identifiable when no two distinct parameter configurations produce the same predicted pattern across all of the model's outputs. Orienting Question: Is there a discussion of the model's structural identifiability?
Filter (E)	Did you evaluate (a part of) this theory / model? ☐ No > out ☐ Yes ☐ ... by using simulations and / or mathematical derivations > S ☐ ... by comparing its predictions against empirical observations > E	Tests by simulation always require the evaluated theory to be formalized. Empirical tests can also be applied to purely narrative theories. Both may apply to the same paper. Orienting Question: <i>Was the theory evaluated not at all, via simulations / mathematical derivations, via empirical data, or both?</i>
ES...	<i>The paper...</i>	
S1	☐ ...specifies any additional auxiliary assumption that had to be added for the purpose of running the simulation (e.g., restriction of parameter ranges, fixed parameters, number of iterations, etc.).	An <i>auxiliary assumption</i> is any assumption or decision not directly implied by the theory but introduced for purposes such as formalization, simulation, prediction, or empirical testing. Orienting Question: <i>Are decisions that are not implied by the theory / model made explicit?</i>
S2	☐ ...provides an assessment of how well the model reproduces the data patterns indicative of the phenomena to be explained.	Such an assessment should always cover whether the model reproduces the data pattern qualitatively but may also cover aspects of quantitative fit, depending on how narrowly the phenomena are defined. Orienting Question: <i>Did the work attempt to reproduce the data pattern indicative of the phenomenon that is to be explained?</i>

S3	☐ ...provides an assessment of the robustness of the simulation results by, for example, using any of the following or similarly appropriate methods: • Varying auxiliary assumptions that could meaningfully affect the results and testing whether the model's success in reproducing the data pattern indicative of the relevant phenomena is dependent on these assumptions. • Exploring the simulation results across a range of justifiable parameter values to characterize the plausible parameter space.	In the best case, the model's success in reproducing the data pattern indicative of the relevant phenomena should not depend on the auxiliary assumptions but only on the core assumptions. Orienting Question: <i>Does the paper implement one of these (or similar) practices to test the robustness of the simulation results?</i>
S4	☐ ...provides an assessment of whether the relevant phenomena arise from the theory's mechanism(s) by, for example, changing (i.e., deleting or replacing) the mechanism(s) to test whether the relevant phenomena fail to emerge.	This concerns the empirical relevance of the theory's mechanism: do the relevant phenomena emerge as a consequence of the core theoretical assumptions, or is it a byproduct of auxiliary assumptions? Altering or removing the mechanism(s) should ideally lead to a failure in reproducing the relevant phenomena. Orienting Question: <i>Does the paper assess whether relevant phenomena emerge from the mechanism(s) postulated by the theory/model?</i>
S5	☐ ...contains an explicit discussion as to what the results of the simulation mean for the theory that was tested (e.g., support, refutation, limitation of scope, adding or subtracting theory elements), also considering the plausibility of auxiliary assumptions that had been added in the process.	A single test often cannot definitively refute a theory because auxiliary assumptions are always involved; the plausibility of these assumptions should therefore be discussed alongside the simulation outcome. Orienting Question: <i>Is there a discussion of what the simulation results imply for the theory that was tested?</i>
E...	The paper...	

E1	☐ ...describes how each element in the theory that is relevant to the current test was operationalized (i.e., measured, estimated, fixed, or manipulated) in an experimental design.	<i>Operationalization</i> is the process of specifying how (some of) the elements of a theory/formal model are measured, manipulated, or estimated to make the theory testable. Orienting Question: <i>Is it clear how each element of the theory was operationalized?</i>
E2	☐ ...makes transparent how predictions were derived from the theory, either by providing a rationale of how the predictions logically follow from the verbal theory, or by formally deriving the predictions from a formal model.	Orienting Question: <i>Is it clear how the tested predictions follow from the theory?</i>
E3	☐ ...provides an assessment of how consistent the theory's / model's predictions are with the observed empirical data.	Orienting Question: <i>Is there an evaluation of how consistent the empirical observations are with the theoretical predictions?</i>
E4	☐ ...provides a discussion of how informative the empirical test was for evaluating the theory. That is, how narrowly the theory constrained the space of plausible observations as opposed to being compatible with a wide range of other plausible observations.	A close fit between theory and data provides strong support only if the theory could have failed, i.e. if it forbids a substantial range of otherwise plausible observations. A theory compatible with almost any result gains little support from a confirming observation. Thus, the more observations the theory rules out, the more informative the test. Orienting Question: <i>Is there a discussion of whether the theory forbids a substantial range of otherwise plausible observations?</i>
E5	☐ ...contains an explicit discussion as to what the empirical results mean for the theory that was tested (e.g., support, refutation, limitation of scope, adding or subtracting theory elements), considering the plausibility of auxiliary assumptions that had been added in the process.	Orienting Question: <i>Is there a discussion of what the empirical results imply for the theory that was tested?</i>

Filter (C)	Did you compare the tested theory/model to other theories/models? ☐ No > out ☐ Yes > C	These items only apply when the paper explicitly compares two or more theories / models.
C...	The paper...	
C1	☐ ...provides a comparison of each theory's / model's predictive accuracy, i.e., how well they reproduce the data pattern indicative of the phenomena of interest.	Ideally, predictive accuracy should be assessed out-of-sample, that is, on a data set that was not used to construct and/or fit the theory/model. Orienting Question: <i>Is there a comparison of the predictive accuracy of all of the compared theories' / models' predictions?</i>
C2	☐ ...provides a comparison of how narrowly each theory / model constrains the space of plausible observations as opposed to being consistent with a wide range of other plausible observations.	Two theories / models may reproduce the target phenomenon equally well but differ in how much else they permit. A theory that predicts the data pattern indicative of the relevant phenomena while ruling out more alternative observations is better supported when its prediction holds compared to a theory that is also consistent with other alternative observations. Orienting Question: <i>Is there a discussion of how narrowly each theory / model constrains the predictive space?</i>

C3	☐ ...provides a comparison of each theory's / model's explanatory breadth, i.e., the range of other relevant phenomena each theory / model can account for.	Two theories / models can account for a different number of phenomena. A theory / model that accounts for more phenomena than another theory / model, has higher explanatory breadth and should be preferred. Note that this point is different from C2, which is about how precisely each of these phenomena is predicted. Orienting Question: <i>Is there a discussion of the explanatory breadth of the theories / models?</i>
C4	☐ ...provides a comparison of each theory's / model's parsimony, i.e., its simplicity in explaining the relevant phenomena.	Theories / models with fewer parameters that nevertheless explain all the phenomena of interest should be preferred over theories / models with more parameters that explain the same number of phenomena. Orienting Question: <i>Is there a discussion of the parsimony of the theories / models?</i>
C5	☐ ...explicitly discusses the trade-off between each theory's / model's predictive accuracy, precision, explanatory breadth, and parsimony.	Being inferior to another theory / model in one regard may be offset by being superior in another. For instance, if a theory / model has one parameter more than another but explains many more phenomena, it may be plausible to prefer the less parsimonious over the more parsimonious theory / model. Orienting Question: <i>Is there a discussion of the trade-off between each theory's / model's characteristics?</i>

References

- Aczel, B., Szasz, B., Sarafoglou, A., Kekecs, Z., Kucharský, Š., Benjamin, D., Chambers, C. D., Fisher, A., Gelman, A., Gernsbacher, M. A., Ioannidis, J. P., Johnson, E., Jonas, K., Koustas, S., Lilienfeld, S. O., Lindsay, D. S., Morey, C. C., Munafò, M., Newell, B. R., ... Wagenmakers, E.-J. (2019). A consensus-based transparency checklist. *Nature Human Behaviour*, 4(1), 4–6.
<https://doi.org/10.1038/s41562-019-0772-6>
- Bogen, J., & Woodward, J. (1988). Saving the Phenomena. *The Philosophical Review*, 97(3), 303.
<https://doi.org/10.2307/2185445>
- Eronen, M. I., & Bringmann, L. F. (2021). The Theory Crisis in Psychology: How to Move Forward. *Perspectives on Psychological Science*, 16(4), 779–788.
<https://doi.org/10.1177/1745691620970586>
- Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic ambiguity in the social sciences. *Social Psychological Bulletin*, 18, e9923. <https://doi.org/10.32872/spb.9923>
- Gärtner, A., Leising, D., Freyer, N., Musfeld, P., Lange, J., & Schönbrodt, F. (2025). Responsible Research Assessment II: A specific proposal for hiring and promotion in psychology. *Meta-Psychology*, 9. <https://doi.org/10.15626/MP.2024.4604>
- Glennan, S. (2017). *The new mechanical philosophy*. Oxford University Press.
- Glöckner, A., Burkhardt, M., Frank, M., Fuenderich, J., Gießing, C., Haim, M., Harper, K., Melinscak, F., Renkewitz, F., Tho Pesch, F., & Schmalz, X. (2026). *Extending Reproducibility to Theory: Theoretical Reproducibility and Precision as Foundations of Cumulative Science*. PsyArXiv.
https://doi.org/10.31234/osf.io/f6tja_v1
- Glöckner, A., & Fiedler, S. (2025). Addressing Psychology's Theory Crisis: Advances in Theory Specification and Theory Building. *Zeitschrift Für Psychologie*, 233(4), 219–223.
<https://doi.org/10.1027/2151-2604/a000608>
- Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good? *Computational Brain & Behavior*, 7(4), 508–522. <https://doi.org/10.1007/s42113-023-00193-2>
- Guest, O., & Martin, A. E. (2021). How Computational Modeling Can Force Theory Building in Psychological Science. *Perspectives on Psychological Science*, 16(4), 789–802.
<https://doi.org/10.1177/1745691620970585>
- Lange, J., Freyer, N., Musfeld, P., Schönbrodt, F., & Leising, D. (2025). A Checklist for Incentivizing and Facilitating Good Theory Building. *Zeitschrift Für Psychologie*, 2151-2604/a000604.
<https://doi.org/10.1027/2151-2604/a000604>
- Machamer, P., Darden, L., & Craver, C. F. (2000). Thinking about Mechanisms. *Philosophy of Science*, 67(1), 1–25. <https://doi.org/10.1086/392759>

- Meehl, P. E. (1990). Appraising and Amending Theories: The Strategy of Lakatosian Defense and Two Principles that Warrant It. *Psychological Inquiry*, 1(2), 108–141.
https://doi.org/10.1207/s15327965pli0102_1
- Muthukrishna, M., & Henrich, J. (2019). A problem in theory. *Nature Human Behaviour*, 3(3), 221–229. <https://doi.org/10.1038/s41562-018-0522-1>
- Oberauer, K., & Lewandowsky, S. (2019). Addressing the theory crisis in psychology. *Psychonomic Bulletin & Review*, 26(5), 1596–1618. <https://doi.org/10.3758/s13423-019-01645-2>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Robinaugh, D. J., Haslbeck, J. M. B., Ryan, O., Fried, E. I., & Waldorp, L. J. (2021). Invisible Hands and Fine Calipers: A Call to Use Formal Theory as a Toolkit for Theory Construction. *Perspectives on Psychological Science*, 16(4), 725–743. <https://doi.org/10.1177/1745691620974697>
- Schönbrodt, F. D., Gärtner, A., Frank, M., Gollwitzer, M., Ihle, M., Mischkowski, D., Phan, L. V., Schmitt, M., Scheel, A. M., Schubert, A.-L., Steinberg, U., & Leising, D. (2025). Responsible Research Assessment I: Implementing DORA and CoARA for hiring and promotion in psychology. *Meta-Psychology*, 9. <https://doi.org/10.15626/MP.2024.4601>
- Schubert, A.-L., Steinhilber, M., Kang, H., & Quintana, D. S. (2025). Improving statistical reporting in psychology. *Communications Psychology*, 3(1), 156. <https://doi.org/10.1038/s44271-025-00356-w>
- Smaldino, P. E. (2020). How to Translate a Verbal Theory Into a Formal Model. *Social Psychology*, 51(4), 207–218. <https://doi.org/10.1027/1864-9335/a000425>
- Tyner, A. H., Abatayo, A. L., Daley, M., Field, S., Fox, N., Haber, N. A., Hahn, K. M., Struhl, M. K., Mawhinney, B., Miske, O., Silverstein, P., Soderberg, C. K., Stankov, T., Abbasi, A., Aberson, C. L., Aczel, B., Adamkovič, M., Albayrak, N., Allen, P. J., ... Errington, T. M. (2026). Investigating the replicability of the social and behavioural sciences. *Nature*, 652(8108), 143–150. <https://doi.org/10.1038/s41586-025-10078-y>
- Van Dongen, N. N. N., Glöckner, A., Musfeld, P., Sarafoglou, A., Borsboom, D., Bringmann, L. F., Eronen, M., Evers, K., Frankenhuys, W., Fried, E. I., Haslbeck, J. M. B., Helmich, M. A., Laskar, P., Van Lissa, C. J., McElreath, R., Nak, J., Oude Maatman, F., Popov, V., Robinaugh, D., ... Yuan, L. (2026). *The state and status of theory in psychological science*. PsyArXiv. https://doi.org/10.31234/osf.io/2fjx4_v2
- Van Dongen, N. N. N., Van Bork, R., Finnemann, A., Haslbeck, J. M. B., Van Der Maas, H. L. J., Robinaugh, D. J., De Ron, J., Sprenger, J., & Borsboom, D. (2025). Productive explanation: A

framework for evaluating explanations in psychological science. *Psychological Review*, 132(2), 311–329. <https://doi.org/10.1037/rev0000479>

Van Rooij, I., & Blokpoel, M. (2020). Formalizing Verbal Theories: A Tutorial by Dialogue. *Social Psychology*, 51(5), 285–298. <https://doi.org/10.1027/1864-9335/a000428>